UNIVERSITY OF MICHIGAN-DEARBORN • SENIOR DESIGN • 2026

MY BETTER LIFE: AI-POWERED BACKEND DATABASE ENGINEERING FOR PERSONAL DEVELOPMENT PLATFORM

PROJECT SUMMARY

My Better Life is an AI-powered life-guidance platform (NOT a habit tracker) that helps users set personal and professional goals, break them into actionable milestovnes, commit to weekly progress cycles, and answer reflective questions for personalized AI recommendations.

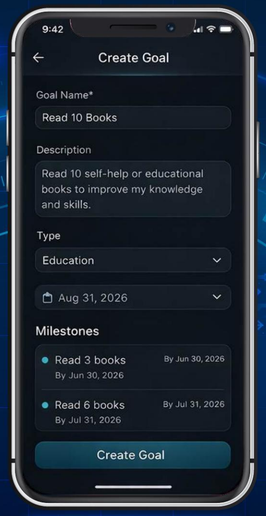
The team engineered the robust Azure Cloud backend database architecture using SQL Server, C#, Blazor Server, and Entity Framework Core. The database has four tables: dbo.Noun, dbo.Type, dbo.POD, dbo.CRUD operations were implemented for all tables using stored procedures for high-performance data access and integrity. Live pie charts visualize live Metric data.



PROGRESS METRICS (Pie Chart)

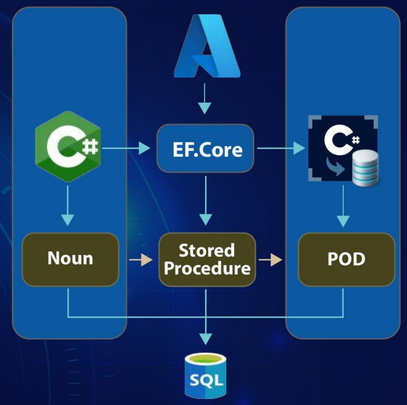


AI REFLECTION QUESTIONS (Chat interface)



GOAL & MILESTONE CREATION

AZURE DATABASE ARCHITECTURE FLOW



PROJECT RESULTS

Developed an Azure-based backend with full CRUD across core tables.

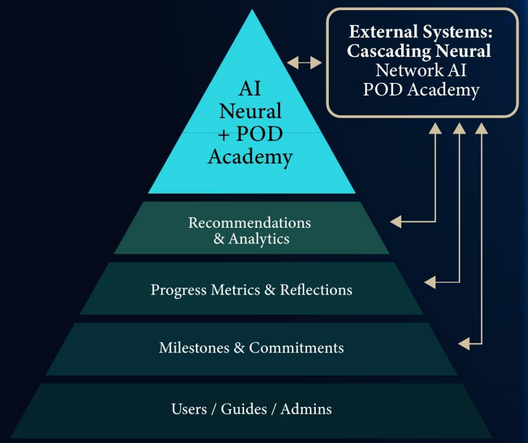
Implemented real-time data visualization using SyncFusion.

Achieved <100ms query latency.

Designed a scalable system supporting AI-driven recommendations and content auditing.

3 DELIVERABLES

PYRAMID MODEL



MARKET RESEARCH & COMPETITIVE ANALYSIS

Feature	Generic Trackers
Structured Reflection	Yes
Cascading Neural Network Integration	Yes
Long-term Goal Setting	No
POD Academy Compliance	No
POD Academy Compliance Auditing	Partial

12 USE CASES

4 DB TABLES

80% SCOPE

<100ms LATENCY

TECHNOLOGIES USED

Azure · EF Core · Microsoft SQL Server · GitHub · Syncfusion

TESTING & VALIDATION

- Data-driven summary: avg 85ms latency
- Data-driven summary: avg 40ms latency
- CRUD reliability verification

DELIVERABLES

•CRUD •Live Charts •Scalable DB

FUTURE VISION & ROADMAP

Phase 1–4

DESIGN ALTERNATIVES CONSIDERED & REJECTED

Team: Michael Berry (Team Lead) and three teammates

Professors: Bruce Maxim, Mahmoud Abou Nasr, Thomas Steiner.